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V300 ZEFIRO

A New Sense of Speed

Passenger Information and Entertainment - VZES SFRS

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Prepared:	LUCA IANDOLO		24/02/2021
Checked:	PIETRO RENZULLI		25/02/2021
Approved:	BRUNO BUSCO		26/02/2021
Released:	GIUSEPPE CARRINO		26/02/2021
	Name	**Signature**	**Date**

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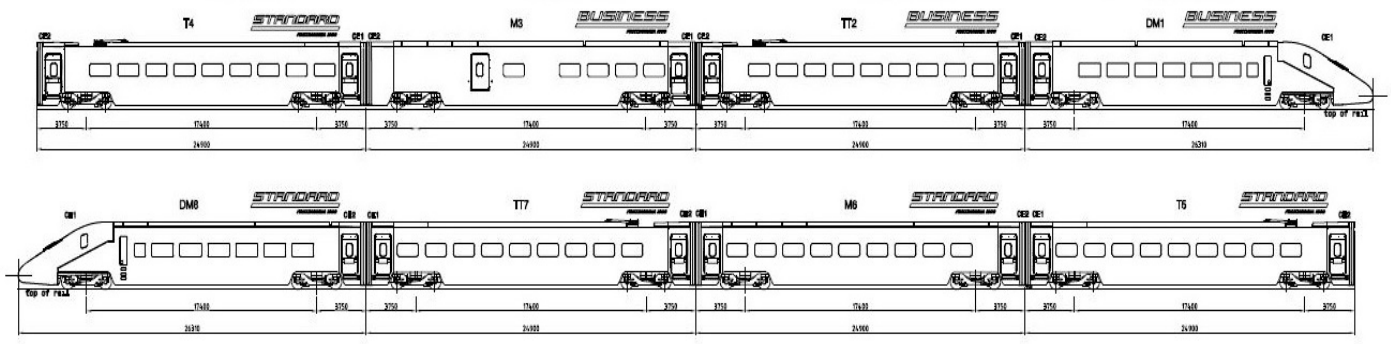
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1 Introduction

This document is a function specification for the control of the *V300ZEFIRO ES PIS* system for Trenitalia customer.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

AP	Access Point
CCU	Coach Control Unit
CD	Circuit Diagram
DMI	Driver Machine Interface
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
OBoE	On Board Equipment (equipment name from system supplier)
GPRS	General Packet Radio System
GPS	Global Position System
GSM	Global System for Mobile Communications
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
HSPA	High Speed Packet Access
IDU	Interface Display Unit
LCD	Liquid Crystal Display
LCU	Local Control Unit
MAR	Mobile Access Router
PDA	Personal Digital Assistant (hand-held computer)
SCU	System Control Unit
TCMS	Train Control and Management System
TSF	Oboe supplier company name
UMTS	Universal Mobile Telecommunications System
Wi-Fi	Wireless Fidelity (IEEE 802.11b wireless networking)

2 System Overview

The architecture is characterized by a Gigabit Electrical Ethernet Backbone.

Moreover, a standard UIC568 bus is foreseen to grant PA communication also in case of multiple faults in the digital backbone and in the relevant circuits.

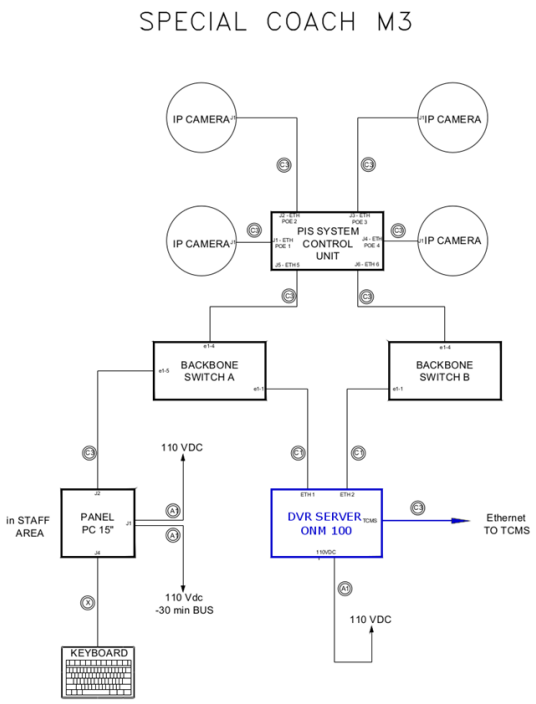
The on-board architecture provides the Internet service through WiFi to navigate via web interface and to access to e-mail services via personal equipment such as Laptop, PDA, Mobile-Phone.
Predisposition for two multimedia servers is also included to distribute audio visual contents on-demand, as film and music, available through the on board WiFi network.

M3 coach is provided with a server with a GSM/GPRS/HSPA module for ground communication with the central database of the railway operator, named Infotren, and a GPS module for the localization of the train position.

In the same coach the DVR/ONM 100 server is employed to implement more functionalities in a single device such as:

- Network management, control and monitoring, based on SNMP
    - Fault management
    - Configuration Management
    - Security Management
- CCTV system recording and management
- TCMS interface management

The PIS System is interfaced to the Train Control Management System to exchange data with other on-board systems. Unlike the previous fleet, this interface is managed by the DVR/ONM 100 Server as shown in the following architectural scheme:



An Infotainment server which runs the TSF application Dove6 Movie, Dove6 On Demand and Eccoti web.

The following list summarizes all the servers of the PIS:

Server name	Location	Function
Infotainment server (DOVE6 Movie Server)	M3	This server hosts the software application that deals with the full management of customer information on the monitors inside the coaches (LCD displays) and on the on-board train portal
Oboe	M3	This device manages the ground communication with the central database of the railway operator
DVR/ONM 100 server	M3	This server implements more functionalities such as network monitoring and management, recording CCTV images and TCMS interface management.
Panel PC	M3	This device is used by the on-board staff to manage the system

Table 1: Servers list

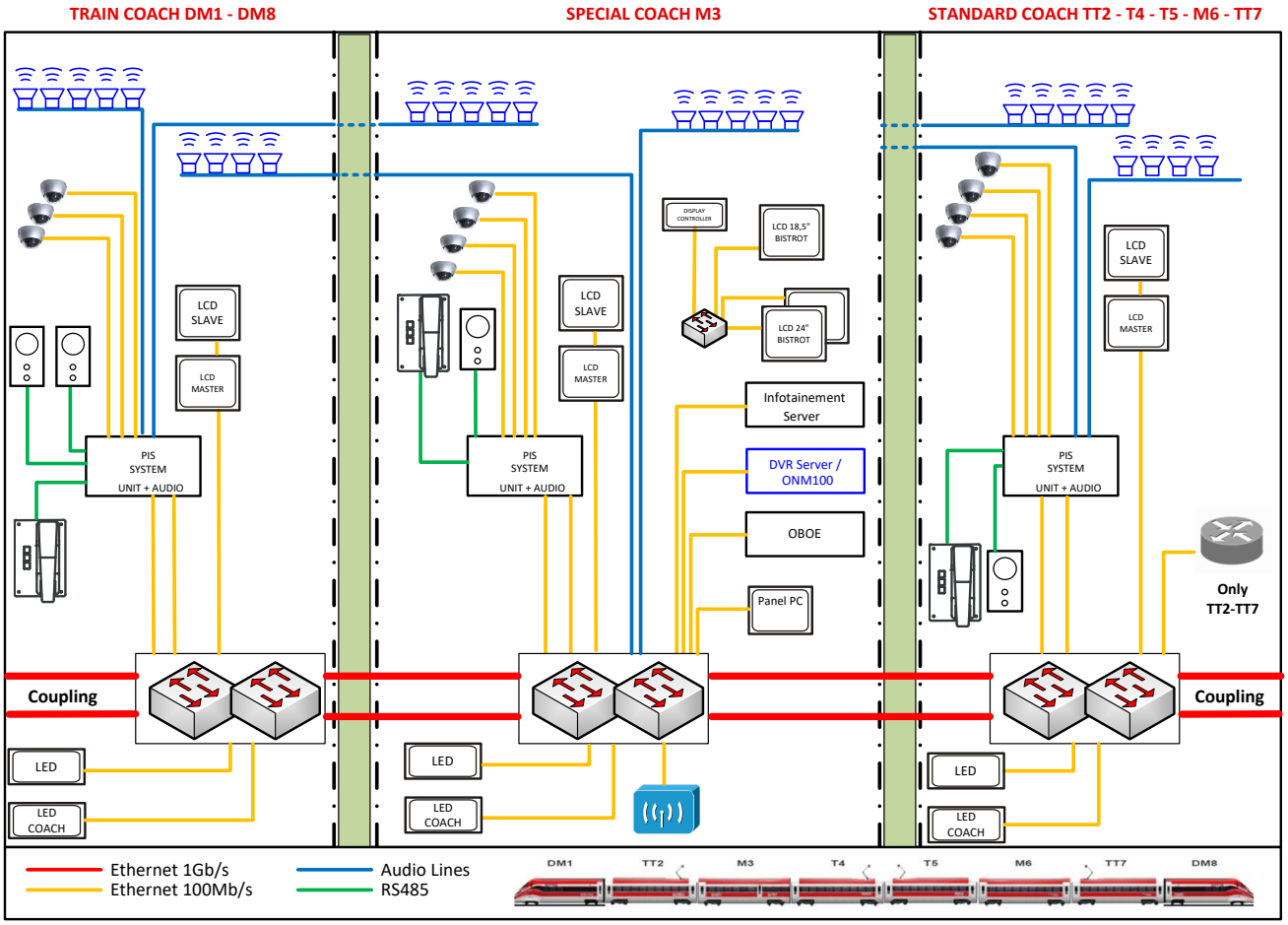
Each coach is equipped with one PIS LCU, while M3 coach is equipped with the PIS SCU.

The train staff area is equipped with a touch screen MMI console with also a keyboard for control of the complete system by the train staff.
Main functionalities are:

- Video surveillance management up to four cameras at the same time, or all cameras in cyclic mode
- Manual Audio and Text announcement management
- Parameters configuration
- System diagnostic
- Intercom staff management

MMI functionality is in the PIS system scope. Refer to RD1 for further detailed.

2.1 System Configuration



2.2 Bistro Area display solution

An independent display system is provided in M3 car to show the current Bistro menu and offers to the passengers. The system is composed of:

- 2 24” LCD display
- 1 18.5” LCD display
- 1 Display Controller
- 1 Ethernet Switch

The display and the controller are connected to an Ethernet Switch that is not connected to the PIS System network, making this network completely independent from the PIS network and the TCMS.

The Display Controller let the Staff members to manage the system and to load the contents to be displayed on the screen using a normal USB Flash Drive. The displays will then show the loaded contents in an infinite loop.

3 Functional requiremen

The requirement for the safe shutdown of the servers has been modified compared to the previous fleet (VZI-14) because the new DVR/ONM100 has an integrated solution to guarantee the safe shutdown that doesn’t need the electromechanical solution previously implemented.

Compared to the previous fleet also new passenger emergency intercoms have been added in the vestibules to comply to the latest TSI as shown in section 3.3.

3.1 General PIS Control

3.1.1 External Interfaces

3.1.1.1 Function Design

The M3 is provided with a server (Oboe) with a GSM/GPRS/HSPA module for ground communication with the central database of the railway operator, named Infotren, and a GPS module for the localization of the train position.

The PIS System is interfaced also to the Train Control Management System (via Ethernet link to the backbone) used to exchange data with other onboard systems, like odometer and diagnostics.
All content, programs, playlists, "spot" events are managed entirely at the central level on ground, as well as the definition of the mission data for each race.

The TCMS shall be able to command the system shut down in advance to the removal of the power supply in order to avoid problem on the Servers (all the servers listed in Table 1).

The TCMS shall also be able to set all the LCD displays in stand-by mode through a command using Ethernet, this is used for example during load reduction, refer to Battery SFRS.

To ensure a graceful shutdown of the servers in Table 1 (except the OBoE and DVR/ONM 100 server), an electromechanical solution implemented by the Conventional Train Control shall be foreseen to send a shutdown signal to the servers as described in RD1.

3.1.1.1.1 Normal condition

The PIS is continuously in contact with the ground system and in this way it can show to passenger up to date information.

3.1.1.1.2 Failure cases

A. Loss of ground communication => PIS shows information not up to date
B. Loss of GPS signal =>position is computed from odometry but after some time the drift could be to big
C. Loss of communication with the TCMS => dead reckoning is not available in case of GPS failure, PIS diagnostic data will be not recorded by the TCMS
D. The TCMS fails to warn with the proper timing the PIS about imminent power off => short term, longer next power up time of the system, long term create damage to the servers

3.1.1.1.3 Failure detection

A. Failure detected by the PIS and communicated to TCMS
B. Failure detected by the PIS and communicated to TCMS
C. Failure detected by the TCMS itself
D. Failure detected by the PIS and communicated to TCMS

3.1.1.1.4 Failure handling

A. Failure isolated and repaired during maintenance
B. Failure isolated and repaired during maintenance
C. Failure isolated and repaired during maintenance

D. No action

3.1.1.2 Requirements

ID	2F_11.Zefiro-Europe.TRS.209
Description	The TCMS shall send a signal to PIS control unit before power down.
Safety level	N/A

ID	2F_11.Zefiro-Europe.TRS.178
Description	The system shall be connected to power supply line controlled via TCMS.
Safety level	N/A

ID	2F_11.Zefiro-Europe.TRS.233
Description	At train power off phase, all the servers shall be capable to perform a graceful shutdown.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_11.Zefiro-Europe.CONCEPT.135	The PIS equipment necessary for the function Public Address and intercom shall be supplied on all the operational mode	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_11.Zefiro-Europe.CONCEPT.395	The Conventional train control shall be able to power off the PIS servers in the M3 car	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_11.Zefiro-Europe.CONCEPT.292	The TCMS shall send to the PIS system though ethernet interface the “PIS server power off” signal 30 seconds before power is cut off to the PIS servers	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_11.Zefiro-Europe.CONCEPT.364	The TCMS shall be able to set all the LCD display in stand by mode through a command using Ethernet.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_11.Zefiro-Europe.CONCEPT.365	The PIS, when commanded by the TCMS, shall set all the LCD displays in stand by mode.	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		SIL 0
2F_11.Zefiro-Europe.CONCEPT.396	The Conventional Train Control shall allow the servers (except the OBoE and DVR/ONM100) to perform a safe shutdown throgh an electromecanical solution when the power supply goes down and a shutdown command has not been issued by the TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		N/A

3.1.2 Internal Interfaces

3.1.2.1 Function Design

The PIS system is managed at single train level by the System Control Unit and the Oboe server in the M3 coach. However, in case of coupled configuration the system shall be interconnected for data exchange as for example camera video images for visualization on train staff console, public address and intercom VoIP data.

3.1.2.1.1 Normal condition

The intercom or public address is based on VoIP protocol also in case of coupled train thanks to the Ethernet connection in the coupler.

3.1.2.1.2 Failure cases

A. Failure of Ethernet connection in the coupler => intercom and PA are realized through the bus UIC 568. It is no more possible to have video images of cameras of the coupled train

3.1.2.1.3 Failure detection

A. Failure detected by the PIS and communicated to the TCMS

3.1.2.1.4 Failure handling

A. Failure isolated and repaired during maintenance

3.1.2.2 Requirements

ID	2F_11.Zefiro-Europe.TRS.175
Description	The PIS system shall support trains operating in multiple.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_11.Zefiro-Europe.CONCEPT.138	The bus UIC 568 interface shall be foreseen in the automatic coupler	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_11.Zefiro-Europe.CONCEPT.298	The dedicated Ethernet network for PIS shall be foreseen in the coupler	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.2 Public Announcements

3.2.1 Function Design

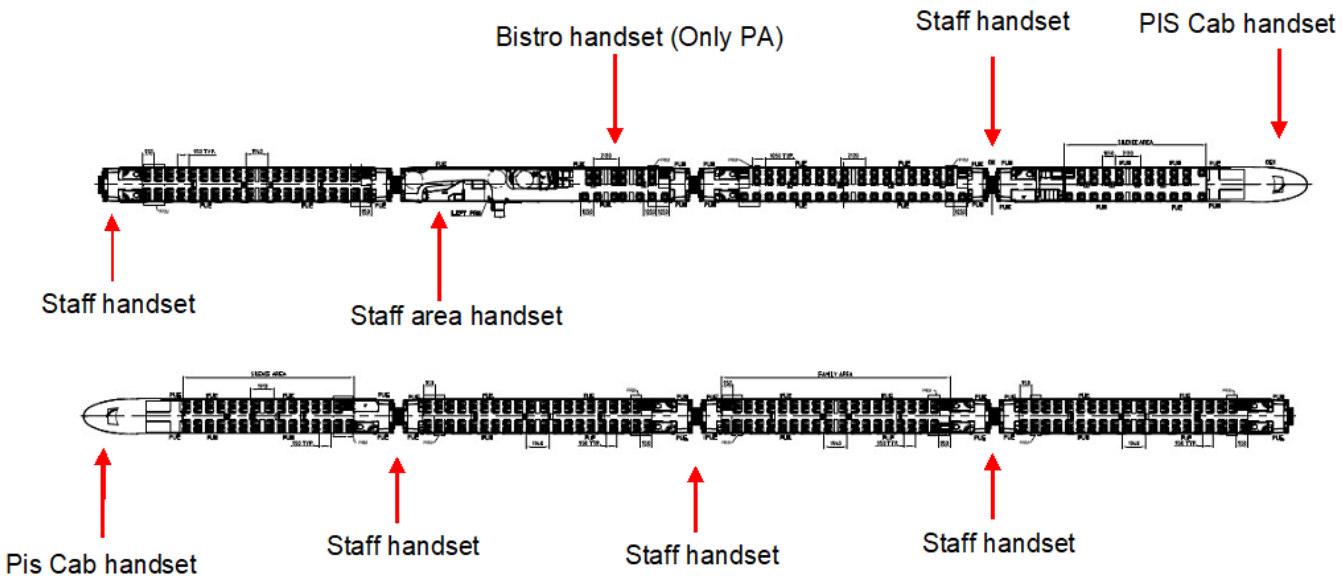
Public announcement and intercommunication are fully integrated with the CAB radio system according to UIC568 and it is realized through VoIP technology provided by the Ethernet backbone and Control Coach Units (CCU). Each CCU includes an UIC568 / VoIP Gateway in order to provide analog to VoIP conversion in standard mode, but also analogue back up mode in case of failure of the Ethernet backbone. In case of Ethernet unavailability, the public address functionality is allowed through UIC bus anyway (public address generated by CAB radio or via PIS handsets). To further increase the reliability each CCU drives the half of loudspeakers installed in the same coach and the half of the adjacent coach, to guarantee the audio announcement in case of one CCU failure.
Intercoms and handset panel are based on serial line technology for call control, thus allowing queuing of simultaneous calls.
Dedicated noise sensors are connected to the control unit in order to measure the noise level of each coach: this level is reported to the local CCU to adjust the volume of loudspeakers.
Automatic audio announcements are managed by the functionality OBOE according to information received from the ground as described in §3.1.1.1.
The system generates automatically, based on train number and current position, all audio announcements related to the trip for the whole train, including delays.
In details messages, managed from Infotren (on ground system), could be:

- Automatic trip announcements (welcome on board, next stops, etc.)
- Recurrent messages
- Spot messages
- Live messages (via GSM)
- Timed messages
- On-demand messages (from Eccoti SW)

The communication system shall be designed in such a manner that it continues to operate at least half (distributed throughout the train) of its loudspeakers in the event of a failure in one of its transmission elements, or another mean shall be available to inform the passengers.
The following components are foreseen for the PA functionality:

	Cab coach	Special coach	Standard coach	Total
Staff area handset	-	1	-	1
Vestibules area handset	-	-	1	5
Cab handset	1	-	-	2
Bar bistrò area handset	-	1	-	1
Loud speakers (Saloon)	6	6	6	48
Loud speakers (vestibules)	1	1	2	13
Loud speakers (Toilet)	2	2	2	16
Noise sensors	To assure the performance	To assure the performance	To assure the performance	To assure the performance

One vestibule for each couple of vestibules except the vestibules adjacent to the special car shall be equipped with a handset providing intercom and PA functionality to the train staff.
In the special car one handset shall then be foreseen in the staff area and a simplified handset in the bistro area (only PA functionality).



The PA system shall reproduce a specific tone when requested by the TCMS through Ethernet.

3.2.1.1 Normal condition

3.2.1.2 Failure cases

3.2.1.3 Failure detection

3.2.1.4 Failure handling

3.2.2 Requirements

ID	2F_11.Zefiro-Europe.TRS.92
Description	The intercom function shall be realized through the Bus UIC 568 to which the cab radio, the PIS handsets of each coach and the passenger emergency device are connected.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_11.Zefiro-Europe.CONCEPT.149	PIS SCU, PIS LCU, PIS handset and PIS emergency intercom device shall be connected to the UIC 568 bus.	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A
2F_11.Zefiro-Europe.CONCEPT.300	The GSM-R cab radio shall be connected to UIC 568 bus	Derived Requirement	4S_10.01.-.Train Radio System		N/A
2F_11.Zefiro-Europe.CONCEPT.366	The TCMS shall command, through Ethernet, the PA to generate a tone.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_11.Zefiro-	The PIS system shall generate a tone on all the	Derived	4S_10.02.00.-		N/A

Europe.CONCEPT.367	coaches when commanded by TCMS.	Requirement	.Passenger Information and Entertainment System - PIES		

3.3 Intercommunication between train staff and between train staff and passengers

3.3.1 Intercommunication from passenger to train personnel

3.3.1.1 Passanger Emergency Communication system

3.3.1.1.1 Function Design

The following components are foreseen for the functionality:

	Cab Coach	Special coach	Standard coach	Total
Passenger emergency intercom	3 (DM8) 3 (DM1)	3	4	29

When the passenger emergency brake handle is activated a signal will be sent to the relative intercom device to establish the communication between passenger and driver. See also BRAKE SFRS RD3. The intercom device will then communicate through RS-485 the information to the LCU. The LCU then communicates the information to the DVR/ONM 100 Server that will communicate the following information to the TCMS through the Ethernet interface: Emergency Brake handle number x (X=1:4 for each car except on Cab Coaches and M1 where x=1:3) on coach y (where y=1:8) has been activated.

3.3.1.1.1.1 Normal condition

In case of activation of the Passenger emergency brake handle, the Passenger emergency intercom forwards a call to the driver. The communication is then established only when the driver accepts the call.

3.3.1.1.1.2 Failure cases

A. Failure of the Passenger emergency intercom => the call is not forwarded to the driver

B. Failure of the brake handle contact => the driver is informed of the Passenger emergency brake activation, but the call is not forwarded to the driver

3.3.1.1.1.3 Failure detection

A. The device functionality is continuously monitored

B. No diagnostic

3.3.1.1.1.4 Failure handling

A. Device to be replaced during maintenance

B. Failure isolated and repaired during maintenance

3.3.1.1.2 Requirements

ID	2F_11.Zefiro-Europe.TRS.130
Description	Pulling down the emergency brake handle shall activate the Emergency Speech Unit.
Safety level	SIL 0

ID	2F_11.Zefiro-Europe.TRS.43
Description	There shall be an interface between the emergency brake handle and the passenger emergency intercom to allow communication between the passenger and the driver.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_11.Zefiro-Europe.CONCEPT.176	The status of the passenger emergency brake handle shall be sent to the PIS intercom device	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_11.Zefiro-Europe.CONCEPT.307	The PIS system shall acquire the status of a contact of the passenger emergency brake handle (PEB handle status) to request to start communication with the driver	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		SIL 1
2F_11.Zefiro-Europe.CONCEPT.310	The PIS shall establish the communication between the driver and the intercom when the driver accept the call	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		SIL 1
2F_11.Zefiro-Europe.CONCEPT.311	The PIS shall inform the passenger that the call has been established	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		SIL 1
2F_11.Zefiro-Europe.CONCEPT.368	The PIS shall communicate through Ethernet to the TCMS the idecntification number of the intercom device that has been operated.	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A
2F_11.Zefiro-Europe.CONCEPT.369	The TCMS shall acquire the identification number of the intercom device that has been operated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.3.1.2 Emergency call from Toilet

3.3.1.2.1 Function Design

In case of activation of the alarm in the toilette for person with reduced mobility (information received from the TCMS through Ethernet) the system shall send an alarm message to the train staff handheld terminal and shall visualize an alarm on the train staff MMI.

3.3.1.2.1.1 Normal condition

When the push button in the toilette is activated a message reach the train staff on the handheld terminal and on the MMI in the staff area

3.3.1.2.1.2 Failure cases

A. Ethernet communication failure between TCMS and PIS => no message is delivered to train staff
B. WLAN failure => no message delivered to train staff handheld terminal

3.3.1.2.1.3 Failure detection

A. Failure detected by the TCMS
B. Failure detected by the PIS and communicated to TCMS

3.3.1.2.1.4 Failure handling

A. Failure isolated and repaired during maintenance
B. Failure isolated and repaired during maintenance

3.3.1.2.2 Requirements

ID	2F_11.Zefiro-Europe.TRS.203
Description	When an Emergency Call Button is pressed the system shall show an alarm on the PIS MMI in the crew cabin. The alarm shall include the position of the emergency call in the train
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_11.Zefiro-Europe.CONCEPT.179	The TCMS shall send through Ethernet the information UNI toilet Alarm activated	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_11.Zefiro-Europe.CONCEPT.318	The PIS shall send an alarm message to the train staff handheld terminal indicating the alarm location when the information UNI toilet Alarm activated is received from the TCMS	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		SIL 0
2F_11.Zefiro-Europe.CONCEPT.319	The PIS shall display a message on the MMI in the staff area indicating the alarm location when the information UNI toilet Alarm activated is received from the TCMS	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		SIL 0

3.3.2 Request from wheelchair user

3.3.2.1 Function Design

Request from external doors
The system shall acquire the status of a pushbutton located on the door dedicated to the access of people with reduced mobility (two doors on the special car) in order to forward the request to the train staff. In case of activation of the pushbutton the system shall send a message to the train staff handheld terminal and shall visualize it on the train staff MMI.
Request from wheel chair position
The system shall acquire the status of one or two pushbuttons located in the wheelchair position. In case of activation of the pushbutton the system shall send a message to the train staff handheld terminal and visualized on the train staff MMI.

3.3.2.1.1 Normal condition

When the push button is activated a message reaches the train staff on the handheld terminal and on the MMI in the staff area.

3.3.2.1.2 Failure cases

A. Pushbutton switch acquisition failure => no message is delivered to train staff
B. WLAN failure => no message delivered to train staff handheld terminal

3.3.2.1.3 Failure detection

A. No diagnostic
B. Failure detected by the PIS and communicated to TCMS

3.3.2.1.4 Failure handling

A. Failure isolated and repaired during maintenance
B. Failure isolated and repaired during maintenance

3.3.2.2 Requirements

ID	2F_11.Zefiro-Europe.TRS.81
Description	The PIS control unit will be interfaced with the door dedicated to the access of people with reduced mobility (two PRM doors on the special car) in order to forward the request to the train staff on the dedicated MMI on the special car.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_11.Zefiro-Europe.CONCEPT.370	There shall be a pushbutton near the PRM doors for requesting access.	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A
2F_11.Zefiro-Europe.CONCEPT.180	Status of PRM access request pushbutton (near PRM doors) shall be sent to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_11.Zefiro-Europe.CONCEPT.326	The status of the pushbutton near the PRM doors (PRM access request) shall be sent to the PIS SCU.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_11.Zefiro-Europe.CONCEPT.325	When received the PRM access request from the TCSM, the PIS shall send a message to the train staff handheld terminal when the input PRM access request is set to HIGH	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A
2F_11.Zefiro-Europe.CONCEPT.329	The PIS shall display a message on the MMI in the staff area when the input PRM access request is set to HIGH	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A

ID	2F_11.Zefiro-Europe.TRS.227
Description	When a PRM help request button in the wheelchair space is pressed the system shall show an alarm on the PIS MMI in the crew cabin. The alarm shall include the position of the emergency call in the train
Safety level	N/A

ID	2F_11.Zefiro-Europe.TRS.228
Description	When a PRM help request button in the wheelchair space is pressed the system shall send an alarm to the train staff handheld terminal. The alarm shall include the position of the emergency call in the train
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_11.Zefiro-Europe.CONCEPT.330	The status of the PRM help request pushbutton (near wheelchair position) shall be sent to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_11.Zefiro-Europe.CONCEPT.371	There shall be one of two pushbuttons near the wheel chair position for requesting help	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A
2F_11.Zefiro-Europe.CONCEPT.331	The status of the pushbutton near the wheel chair position (PRM help request) shall be sent to the PIS SCU.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_11.Zefiro-Europe.CONCEPT.332	When received the PRM help request from the TCMS the PIS shall send a message to the train staff handheld terminal when the input PRM help request is set to HIGH	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A

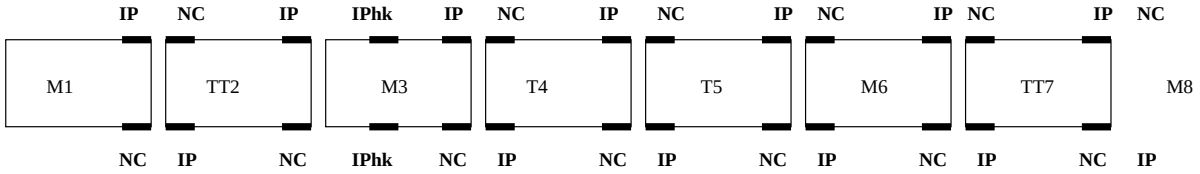
2F_11.Zefiro-Europe.CONCEPT.333	The PIS shall display a message on the MMI in the staff area when the input PRM help request is set to HIGH	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A

3.4 Displays

3.4.1 Exterior Display

3.4.1.1 Function Design

The external displays will be managed by the PIS according to database information received from ground as described in §3.1.1.1.
The external displays are foreseen on external doors according to the following scheme:



IP = Full external display

IPhk = Full external display on HK coach

NC =Coach number display

Full external display
The side display will have a matrix made of 132x48 LEDs with a dot pitch of 4 mm.
The matrix shall be virtually split into three sectors:

- Sector A: matrix 50x21 for the Coach Number and Coach Pictogram (fixed).
- Sector B: matrix 80x21 for Train number (fixed).
- Sector C: matrix 132x26 for Destination and Stopping Stations. The former will be displayed with characters 12 rows high, the latter with characters 8 rows high. Sector C will have scrolling capability.

The LEDs are high intensity type with automatic brightness adjustment according to the ambient light and they are switched off at train speed greater than a predefined speed.
The sign is driven by means of Ethernet (M12) connection. Addressing of each display inside the coach is configured by using the Ethernet port. This feature shall allow the addressing of the single sign in case of diagnostic information request. The availability of diagnostic functions allows remote SW upgrade and maintainability.
In case of the door is out service (info provided by TCMS), the display shows the “ouf of service” indication instead of the coach number.
Coach number display
The side display will have a matrix made of 51 x 18 LEDs with a dot pitch of 4 mm.
The matrix will be used to show the coach number and the coach icon, same as sector A of side LED display.
Also this sign, driven by means of Ethernet connection, will be switched off at train speed greater than a predefined speed.
In case of the door is out service (info provided by TCMS), the display shows the “out of service” indication instead of the coach number.

3.4.1.1.1 Normal condition

The PIS is continuously in contact with the ground system and in this way, it can show to passenger up to date information on LED display.

3.4.1.1.2 Failure cases

A. Loss of ground communication => PIS shows information not up to date
B. Failure of a LED display => people can look at further display

3.4.1.1.3 Failure detection

A. Failure detected by the PIS and communicated to TCMS
B. Failure clearly evident to train staff

3.4.1.1.4 Failure handling

A. Failure isolated and repaired during maintenance
B. Device replaced during maintenance

3.4.1.2 Requirements

ID	2F_11.Zefiro-Europe.TRS.31
Description	Automatic management of internal and external displays by the OBoE system
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_11.Zefiro-Europe.CONCEPT.199	The TCMS shall send to the PIS thorugh Ethernet the individual External door status (ok / isolated).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_11.Zefiro-Europe.CONCEPT.342	The PIS, in case of one or more doors out of service, shall show on the external display of the relevant door, the door isolated status.	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A

ID	2F_11.Zefiro-Europe.TRS.232
Description	The coach number for each coach shall be derived, also in the case of multiple unit, and displayed on the external displays.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_11.Zefiro-Europe.CONCEPT.341	The PIS shall derive the coach number for each coach, also in case of multiple unit, based on the information trip number and coupled/uncoupled	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A
2F_11.Zefiro-Europe.CONCEPT.345	The PIS shall show on each external display the coach number.	Derived Requirement	4S_10.02.00.-.Passenger Information and Entertainment System - PIES		N/A
2F_11.Zefiro-Europe.CONCEPT.340	The TCMS shall send thorugh Ethernet the information Coupled/uncoupled to the PIS	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

The trip number is received by the PIS system from the ground system.

3.4.2 Interior Display

3.4.2.1 Function Design

The internal displays will be managed by the PIS according to information received from ground as described in §3.1.1.1 except for the displays, located in the Bistro area of M3 coach, which have a dedicated controller.

3.4.2.1.1 Normal condition

The PIS is continuously in contact with the ground system and in this way, it can show to passenger up to date information on LCD display.

3.4.2.1.2 Failure cases

A. Loss of ground communication => PIS shows information not up to date
B. Failure of an LCD display => people can look at further display

3.4.2.1.3 Failure detection

A. Failure detected by the PIS and communicated to TCMS
B. Failure clearly evident to train staff

3.4.2.1.4 Failure handling

A. Failure isolated and repaired during maintenance
B. Device replaced during maintenance

3.4.2.2 Requirements

The toilet directly manages the indicator panel installed in the compartment.
The TCMS sends to the PIS the following information for the proper working of the OBoE functionality:

Nr	Description	Type	Derived to	User Interface	SIL
2F_11.Zefiro-Europe.CONCEPT.404	TCMS shall send train speed to PIS	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_11.Zefiro-Europe.CONCEPT.405	TCMS shall send Door status (released, not released) to PIS	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.5 Events

3.5.1 Function Design

All the system diagnostic shall be sent to the TCMS through Ethernet.
Refer to RD2 for the complete list signal.

3.5.2 Requirements

ID	2F_11.Zefiro-Europe.TRS.224
Description	Diagnostics from the supplier shall be stored in the train diagnostic system.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_11.Zefiro-Europe.CONCEPT.245	Diagnostics from the supplier shall be stored in the train diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_11.Zefiro-Europe.CONCEPT.28	PEB handle status	L	2F_11.Zefiro-Europe.CONCEPT.176

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_11.Zefiro-Europe.CONCEPT.29			

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_11.Zefiro-Europe.CONCEPT.30	PRM access request	L	2F_11.Zefiro-Europe.CONCEPT.180
2F_11.Zefiro-Europe.CONCEPT.354	PRM help request	L	2F_11.Zefiro-Europe.CONCEPT.330

Outputs:

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_11.Zefiro-Europe.CONCEPT.32	Emergency Brake handle number		2F_11.Zefiro-Europe.CONCEPT.368

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_11.Zefiro-	PIS server power off	L	2F_11.Zefiro-Europe.CONCEPT.292

Europe.CONCEPT.33			
2F_11.Zefiro-Europe.CONCEPT.356	UNI toilet Alarm	L	2F_11.Zefiro-Europe.CONCEPT.179
2F_11.Zefiro-Europe.CONCEPT.372	PIS LCD to standby		2F_11.Zefiro-Europe.CONCEPT.364
2F_11.Zefiro-Europe.CONCEPT.373	PA tone command		2F_11.Zefiro-Europe.CONCEPT.366
2F_11.Zefiro-Europe.CONCEPT.357	PRM access request	L	2F_11.Zefiro-Europe.CONCEPT.326
2F_11.Zefiro-Europe.CONCEPT.362	PRM help request	L	2F_11.Zefiro-Europe.CONCEPT.331
2F_11.Zefiro-Europe.CONCEPT.406	Train speed	A	2F_11.Zefiro-Europe.CONCEPT.404
2F_11.Zefiro-Europe.CONCEPT.360	Door out of service (door ok, door isolated).	L	2F_11.Zefiro-Europe.CONCEPT.199
2F_11.Zefiro-Europe.CONCEPT.407	Door status (released, not released)	L	2F_11.Zefiro-Europe.CONCEPT.405
2F_11.Zefiro-Europe.CONCEPT.359	Coupled/Uncoupled	L	2F_11.Zefiro-Europe.CONCEPT.340

5 References

RD	Code	Title	Source	Revision
RD1	S20056.03.1005TR	Technical Specification - PIS ETR1000 ES	Leonardo	01.00
RD2	3EST000225-7439	Interface Control Document (ICD): PIS CCTV V300ZERIFO	BT	00
RD3	3ECP413922-0502	BRAKE SFRS	BT	Latest
RD4	3EST000225-7908	Battery SFRS	BT	Latest